

# Technical Data Sheet: KNSB Propellant Grain Fabrication Log

**Author:** Bencollins Mureithi

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## Abstract

This technical document details the empirical manufacturing logs, thermal adjustments, and final physical metrics of KNSB propellant Batch 02 processed on Monday, 6th July 2026. This batch achieved an operational saturation point at a dry fuel mass of 591.0 g, requiring a proportional oxidizer integration of 1098.0 g. Following successful casting and cooling protocols, this specific grain has been designated as the formal experimental control baseline to support direct comparative performance tracking in upcoming static test fires.

## Section 1: Achieved Batch Specifications Checklist

The real-world metrics, scale restrictions, and physical dimensions captured during the casting run are organized in Table II.

**Table II: Achieved Specifications and Physical Metrics**

| Parameter / Identifier          | Target Specification         | Achieved Metric                  |
|---------------------------------|------------------------------|----------------------------------|
| Experimental Role               | Static Test Baseline Control | Established Control Grain        |
| Propellant Formulation          | KNSB (65:35)                 | 65.01% Oxidizer : 34.99% Fuel    |
| Initial Fuel Mass (Sorbitol 70) | 800.0 g                      | 800.0 g                          |
| Final Processed Fuel            | 560.0 g (Target)             | 591.0 g (Operational Saturation) |

|                                                 |                       |                                     |
|-------------------------------------------------|-----------------------|-------------------------------------|
| <b>Mass</b>                                     |                       |                                     |
| <b>Measured Oxidizer Mass (KNO<sub>3</sub>)</b> | 1097.57 g (Calc)      | 1098.0 g (1g balance resolution)    |
| <b>Slurry Blending Temperature</b>              | 90°C (30 min)         | 90°C (Thermal Stability Maintained) |
| <b>Actual Casting Temperature</b>               | 60°C                  | 60°C (Poured Into Liner)            |
| <b>Liner Tare Weight</b>                        | Standard Baseline     | 145.0 g                             |
| <b>Measured Propellant Net Mass</b>             | Calculated Yield      | 1205.0 g (After subtraction)        |
| <b>Total Assembly Gross Mass</b>                | Completed System      | 1305.0 g                            |
| <b>Grain Outer Radius</b>                       | 8.60 cm               | 8.60 cm                             |
| <b>Grain Internal Core Radius</b>               | 3.55 cm               | 3.55 cm                             |
| <b>Grain Column Length</b>                      | 14.50 cm              | 14.50 cm                            |
| <b>Density</b>                                  | 1.84g/cm <sup>3</sup> | 1.715 g/cm <sup>3</sup>             |

